

Software Requirements Specification (SRS)

Project Mansakha - SIH 2026

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Project Mansakha: AI-Based Dynamic Mental Health Monitoring & Distress Prediction System

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for **Mansakha**, an AI-driven platform designed to continuously monitor and predict psychological distress among victims of atrocities. This SRS details the functional and non-functional requirements, system architecture, user personas, and external integrations necessary to fulfill the SIH-2026 Problem Statement.

1.2 Scope

Mansakha serves beneficiaries receiving relief, compensation, rehabilitation, and protection under the provisions of the Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Act, 1989. The system monitors victims through the investigation, trial, rehabilitation, and compensation processes, predicting escalation in psychological distress and triggering timely interventions by designated authorities and counsellors.

1.3 Definitions and Acronyms

NHAA: National Helpline Against Atrocities (14566). **PII**: Personally Identifiable Information. **IVRS**: Interactive Voice Response System. **SRS**: Software Requirements

Specification. **LLM**: Large Language Model (e.g., Gemma, Ollama).

2. Overall Description

2.1 Product Perspective

Mansakha is a hybrid, distributed system comprising three core components: 1. **Victim Mobile Application (React Native)**: A multilingual interface for victims to access wellness tools, chat with an AI, log journals, and trigger SOS alerts. 2. **Staff Web Dashboard (React.js)**: A role-based command center for Counsellors, Data Operators, Administrators, and Ministry officials to manage cases and view analytics. 3. **Core Backend (Node.js/Express + PostgreSQL)**: A centralized API that handles data routing, AI orchestration, database management (via Supabase), and background dispatch workers.

2.2 Target Audience & Accessibility

The primary beneficiaries are marginalized individuals, including rural populations and families affected by caste-based violence. Because digital literacy and smartphone ownership vary, the system **MUST** support multi-channel interactions, explicitly including low-tech channels (IVRS and SMS) alongside the smartphone app.

2.3 User Personas (Role-Based Architecture)

The system supports exactly 7 distinct personas: 1. **Victim**: The end-user. Interacts with the system via App, SMS, or IVRS. 2. **Data Operator**: Staff responsible for initial data intake and case registration from the NHAA helpline. 3. **Counsellor**: Mental health professionals assigned to specific victims to monitor distress, chat, and log interventions. 4. **District Admin**: Manages counsellor assignments and localized analytics at the district level. 5. **State Admin**: Oversees operations, analytics, and resource allocation across an entire state. 6. **National Admin**: Oversees nationwide operations and state-level metrics. 7. **Ministry Official**: Top-level executive access for nationwide heatmaps, systemic audits, and policy enforcement (cannot view individual PII).

3. Functional Requirements

3.1 Data Intake & Registration

FR-1.1 (NHAA Integration): The system shall integrate with the NHAA (14566) and Integrated Portal via secure webhooks to automatically fetch new case dockets. *(Note: Currently a simulated dummy feature; requires implementation).* **FR-1.2 (Manual Registration):** Data Operators shall have a dedicated web interface to manually register victims, auto-generating a secure docket number and temporary password (*Victim123*).

3.2 Multi-Channel Victim Interaction

FR-2.1 (Mobile App): Victims shall have a smartphone app to complete daily check-ins, write trauma journals, and chat with an AI companion. **FR-2.2 (IVRS Calls):** The system shall utilize Exotel (or similar) to make automated, periodic phone calls to victims on feature phones, logging keypad responses. *(Note: Currently a non-functioning stub).* **FR-2.3 (SMS Prompts):** The system shall utilize Twilio (or similar) to send periodic SMS check-ins to victims without internet access. *(Note: Currently a non-functioning stub).* **FR-2.4 (Multilingual):** All victim-facing text and voice channels must support local Indian languages via translation layers.

3.3 AI Distress Scoring & Predictive Analytics

FR-3.1 (Text Analysis): The AI shall analyze chat logs and journals to compute a Dynamic Distress Score (0-100) based on Sentiment, Emotion (fear/sadness), and Engagement patterns. **FR-3.2 (Voice Analytics):** The system shall analyze audio from IVRS calls or App voice notes for vocal stress indicators. *(Note: Currently deferred/missing).* **FR-3.3 (Predictive Escalation):** The AI must project historical distress trends to preemptively flag a victim *before* they reach a critical threshold. **FR-3.4 (Automated Interventions):** The AI shall suggest appropriate interventions (e.g., Medical, Relocation, Legal Aid) based on the context of the victim's responses.

3.4 Alerts & Case Management

FR-4.1 (Threshold Alerts): If a victim's Distress Score crosses 55 (High) or 80 (Critical), the system shall instantly trigger an alert to their assigned Counsellor. **FR-**

4.2 (Auto-Assignment): High-risk unassigned cases shall be automatically routed to the Counsellor in that jurisdiction with the lowest active workload. **FR-4.3**

(Intervention Logging): Counsellors must be able to log actions taken, overriding or confirming the AI's suggested interventions.

3.5 Analytics Dashboards

FR-5.1 (Hierarchical Views): Dashboards shall strictly filter data based on the viewer's jurisdiction (District vs. State vs. National). **FR-5.2 (Ministry Heatmap):** The Ministry shall have access to geographic heatmaps highlighting high-risk zones and emerging syndicates (organized intimidation).

4. Non-Functional Requirements

4.1 Privacy & Data Security (CRITICAL)

NFR-1.1 (PII Redaction): Before any victim text is sent to a third-party LLM (e.g., Google Gemma via API) for analysis, all Personally Identifiable Information (Names, Locations, Phone Numbers) MUST be scrubbed or anonymized. (Note: Currently missing, massive vulnerability). **NFR-1.2 (Row-Level Security):** The PostgreSQL database must utilize strict Row-Level Security (RLS) to ensure Staff can only query victims within their authorized jurisdiction. **NFR-1.3 (Ministry Blindness):** Top-level Ministry roles must be technically restricted from viewing individual victim names or contact details, seeing only aggregated statistics.

4.2 Explainable AI

NFR-2.1 (Transparency): Every AI-generated distress score must be accompanied by a human-readable explanation (e.g., "Score is 85 because the victim explicitly mentioned a recurring threat from their neighbor"). This explanation MUST be visible to the Counsellor in the UI.

4.3 Scalability & Performance

NFR-3.1 (Asynchronous Processing): AI analysis and heavy SMS/IVRS dispatching must be handled via background workers (e.g., `dispatchWorker.js`) to prevent blocking the main API thread. **NFR-3.2 (Offline Fallback):** The mobile application shall utilize local AI models (e.g., Ollama via `useSpeechToText.js`) where possible to function in low-connectivity rural areas.

5. System Interfaces & Database Design

5.1 Database Schema Overview

`roles` / `officials` : Manages staff access, linking them to specific jurisdictions.
`victims` / `victim_identity` : Separates case metadata from highly sensitive PII.
`interactions` / `interaction_signals` : Stores raw transcripts (in Supabase Storage) and the mathematical components of the distress score. `distress_scores` : Stores the final computed score, risk level, and AI explanation.

5.2 External APIs

Google Gemma: For NLP, Sentiment Analysis, and predictive insights. **Exotel / Twilio:** For executing IVRS calls and SMS blasts.

- **NHAA (14566) Webhooks:** For ingesting live case dockets.